

Dheeraj Baghel

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Summary

Backend and Data Engineer with hands-on experience in Python, PySpark, and AWS/GCP, specializing in distributed scraping systems, ETL pipelines, and scalable backend APIs. Proficient in Apache Kafka, Airflow, FastAPI, and LLM application development using LangChain and LangGraph.

Technical Skills

Languages	Python, SQL, JavaScript, C/C++
Frameworks	FastAPI, Flask, PySpark, Pandas, Scrapy, BeautifulSoup, LangChain, LangGraph, LangSmith
Cloud	AWS (EC2, S3, Lambda, Kinesis, Glue, ECS, Redshift), GCP (BigQuery, GCS, Dataflow, Cloud Run, Pub/Sub)
Data & Orchestration	Apache Kafka, Apache Airflow, Apache Hive, Hadoop, Databricks
Tools & OS	Docker, Git, Charles Proxy, Frida, HTTP Toolkit, Playwright, Linux

Experience

MfilterIt Feb 2025 – Present

Software Development Engineer

- Design and operate scalable web and mobile application scraping systems, including Android app reverse engineering using Frida, Charles Proxy, and HTTP Toolkit on Android emulators.
- Build and maintain PySpark-based ETL pipelines processing high-volume scraped data across AWS (Lambda, S3, ECS, Kinesis) and GCP (BigQuery, GCS) for serverless and containerized workloads.
- Develop and expose backend REST APIs using FastAPI and Flask to serve processed data to downstream consumers.

Six Dee Netad Solutions Pvt Ltd April 2024 – August 2024

R&D Trainee

- Built an automation tool to detect and block fraudulent IP traffic in Bing Ads campaigns, reducing manual monitoring efforts by **70%** and improving invalid traffic (IVT) prevention accuracy.
- Automated campaign performance report generation and distribution using Python and Pandas, streamlining analytics workflows.

DEI Summer Research Fellowship June 2023 – November 2023

Research Intern

- Optimized disaster relief vehicle routing using Genetic Algorithms and K-means clustering, achieving measurable gains in solution quality and computational efficiency for TSP-class problems.

Projects

E-Commerce Data Intelligence Pipeline June 2025 – Present

MfilterIt | Software Development Engineer

- Built distributed scrapers on **AWS Lambda** and **EC2** processing **20M+ URLs in under 4 hours** across Blinkit, Zepto, Instamart, Flipkart, Amazon, and Noon; reverse-engineered mobile app APIs using **Frida, Charles Proxy, and HTTP Toolkit**.
- Designed a **Kafka**-based streaming layer where Lambda scrapers publish raw records; a batch consumer aggregates **5,000–7,000 IDs** per cycle and upserts deduplicated records into the database using fault-tolerant merge logic with retry handling.
- Built a multi-stage **PySpark** ETL pipeline — raw → cleaned (validation, null-handling, deduplication) → staging → transformed — computing ecommerce KPIs including price trends, stock velocity, and discount rates across **20M+ daily records**.
- Merged transformed ecommerce data with campaign datasets to surface performance insights; exposed aggregated metrics and campaign analytics to client teams via a **FastAPI** dashboard with role-based access control.

Weapon-Target Assignment Optimization October 2024 – May 2025

DayalBagh Educational Institute, Agra

- Developed a metaheuristic optimization framework for the Static Weapon-Target Assignment Problem (SWTAP) using a Biased Random-Key Genetic Algorithm (BRKGA) with tailored crossover and mutation operators, achieving computationally efficient constraint satisfaction in C/C++.

Publications

- “The Mobile Healthcare Vehicle Traveling Salesman Problem with Constant Budget Constraint: Heuristics and Hybrid Evolutionary Algorithms” — under review, *Journal of the Institution of Engineers (India) Series B*, Springer.

Education

DayalBagh Educational Institute, Agra	2021–2025
<i>B.Tech in Electrical Engineering, Specialization: Computer Science</i>	<i>8.8 CGPA (Top 10%)</i>
REI Intermediate College, Agra	2020–2021
<i>Senior Secondary (PCM)</i>	<i>97.2%</i>
REI Intermediate College, Agra	2018–2019
<i>High School</i>	<i>95.5%</i>